

Optimization of Synchrotron-Based Photoneutron Sources for Subcritical Systems

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ABSTRACT

This work investigates the feasibility of producing high-intensity photoneutron fluxes from synchrotron radiation as an alternative to conventional linear accelerator sources. A detailed Monte Carlo sensitivity study was performed using a development branch of OpenMC that incorporates photonuclear physics capabilities. A representative synchrotron beamline was modeled to quantify the photoneutron yield across a range of target materials, including high-Z (U, W, Ta) and low-Z (Be, D) converters, as well as a composite target (UO₂ + D₂O + BeO). The analysis examined the influence of target geometry (specifically, length and transversal size) on the total neutron yield, accounting for photon self-shielding and off-axis scattering effects. Results show that uranium-based targets provide the highest yield due to combined (γ, n), ($\gamma, 2n$), and photofission reactions, with an optimal thickness around 10–15 cm. Owing to their substantially higher available beam power (up to 200 kW per beamline), synchrotron-driven photon sources can produce absolute neutron fluxes exceeding 1×10^{15} n/s, outperforming existing LINAC-based systems by a factor 2-3. These results establish synchrotron-driven photoneutron sources as a promising candidate for next-generation accelerator-driven systems and other high-flux neutron applications.

Keywords: Photonuclear reactions, Synchrotron radiation, ADS, Neutron sources, Monte Carlo

1. INTRODUCTION

Accelerator-driven neutron sources (ADNS) represent a key technology with potential applications spanning from materials science, neutron imaging, medical isotopes production, Boron Neutron Capture Therapy (BNCT), and power generation [1, 2, 3]. Within the latter framework, neutron sources may be adopted to drive subcritical reactor cores (known as Accelerator Driven Systems or ADS), which provide a promising approach for both energy production and transmutation of long-lived nuclear waste [4, 5], being also a potentially viable alternative to fast breeders and fusion reactors [6].

The most popular and powerful ADNS is represented by spallation sources, which adopts high-energy protons to generate high neutron yields per incident particle, reaching average intensity values up to 1×10^{18} n/s [1]. The spallation technology requires large efforts in the target management, namely ad-hoc cooling system or design of liquid targets, able to sustain the deposited proton energy. Ridikas *et al.* [7] showed that for neutron fluxes below $\sim 1 \times 10^{16}$ n/s, the electron-driven photoneutron source is the most convenient way of producing neutrons from accelerators. In this method, a high-power

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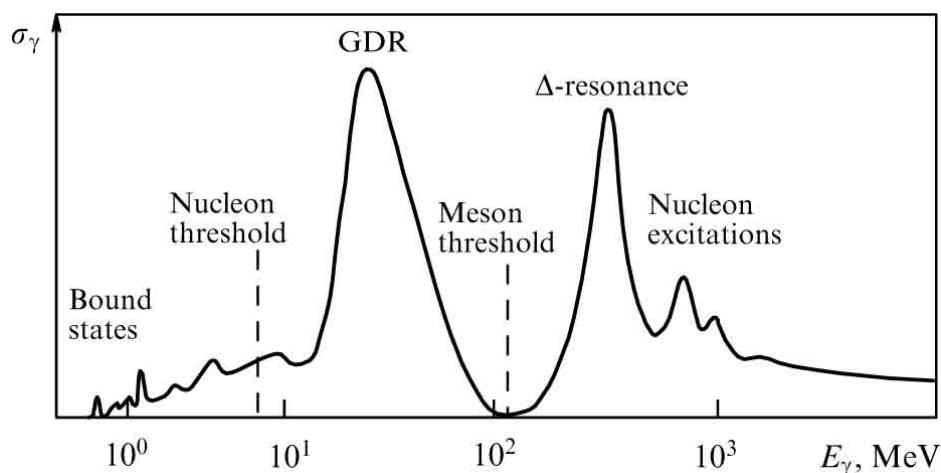


Figure 1. Schematic representation of the absorption photon cross section by atomic nuclei and free nucleons on the photon energy.

electron linear accelerator (LINAC) impinges on a high-Z target (e.g., tungsten or tantalum) to generate bremsstrahlung photons, which produce photoneutrons through the Giant Dipole Resonance (GDR) [8] (details in Figure 1). Compared to spallation sources, this technology notably decreases the thermal load on the target, thereby reducing the effort required for its management and increasing its feasibility. Another improvement is that spallation targets would need to be replaced frequently (every ~ 60 days), while photoneutron targets can last for years without concern [1].

The photoneutron yield (i.e., the number of neutrons produced per second by photon interactions) depends primarily on two factors. The first is the maximum electron energy in the accelerator: higher-energy electrons generate a larger flux of photons in the Giant Dipole Resonance (GDR) region. The second is the *beam power*, which determines how many electrons per second reach the conversion target. Beam power plays a critical role in the design of LINACs because, as in spallation sources, it strongly affects target thermal management. As previously stated, the electron beam impinges on a high-Z target to produce bremsstrahlung photons. This process causes significant heating: for example, at an electron energy of 100 MeV, only about 40% of the beam power is carried away by bremsstrahlung photons, while the remaining 60% is deposited directly in the target as heat. For this reason, a beam power of around 100 kW is generally considered the realistic upper limit for linac-based photoneutron sources [2, 4], as illustrated by facilities such as NSC KIPT [5]. Under these constraints, the maximum achievable photoneutron yield for LINAC-driven systems is $\leq 1 \times 10^{14}$ n/s [1, 9].

To overcome current power limitations and achieve the neutron fluxes required for next-generation ADS and other high-flux applications (including medical uses), this work explores an alternative driver: synchrotron radiation. The use of a large-scale synchrotron as a photon source is complementary to current research trends, which are predominantly oriented toward compact sources [1, 10].

In a synchrotron, electrons circulate along a closed ring and are bent by magnetic fields; the resulting deceleration causes the emission of intense photon beams tangentially to the ring. This configuration offers several advantages:

- **Multiplexing capability:** a sufficiently large synchrotron can operate multiple beamlines simultaneously (up to 40-50). Several photoneutron substations may be installed around the ring, each providing an independent photon beam and serving distinct applications in parallel.
- **High beam power per line:** each beamline can deliver photon beam powers of 200 kW or more, significantly higher than what is feasible with conventional LINAC-based systems, making synchrotrons particularly attractive for high-intensity neutron production.

- **Simplified target management:** photon production in synchrotrons does not rely on a material converter. Consequently, the energy lost by electrons is fully carried away by photons, unlike in LINACs where only a fraction of the beam power emerges as bremsstrahlung. The only thermal load to manage is therefore the neutron-production target itself.
- **High power efficiency:** the electrical power required to re-accelerate electrons after each bending segment is relatively low, with overall efficiencies around $\sim 50\%$.

For sake of simplicity, the remainder of this work will assess the capabilities of a single synchrotron beamline in terms of the maximum realistic neutron yield achievable.

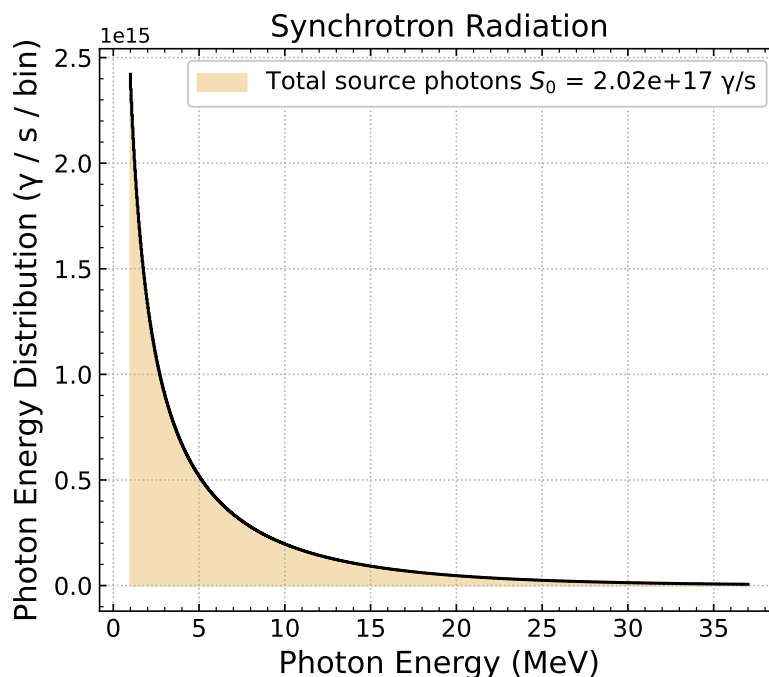


Figure 2. Energy distribution of synchrotron radiation.

This paper presents a simulation sensitivity study using an unofficial branch of the OpenMC Monte Carlo code, which incorporates photonuclear physics capabilities [11]. A realistic synchrotron source is modelled, and its performance in terms of photoneutron yield is evaluated across a comprehensive set of target materials, including conventional high-Z targets (Uranium, Tantalum, and Tungsten) [12, 4] and low-Z, low-threshold targets (Beryllium and Deuterium) [13, 10, 14].

It must be acknowledged that all simulations performed for this work are affected by known biases and inconsistencies in standard photonuclear data libraries, as highlighted in recent benchmark studies [15, 10]. These issues may influence the absolute photoneutron yield by up to a factor of two. Nevertheless, the comparative methodology adopted in this work remains valid for assessing the relative performance of different target configurations, pending the release of corrected data libraries. The ultimate aim of this study is to demonstrate that the synchrotron-driven photon source proposed by this work can substantially outperform LINAC-driven systems in terms of total beam power and neutron yield, establishing a credible pathway toward high-power, multi-purpose neutron sources.

2. METHODOLOGY

The simulation geometry was designed to model a single beamline from a large synchrotron. The synchrotron considered is a 32 GeV ring with a circulating current of 15 mA, in which the photon beam is

generated by a short 15 T, 5 cm long dipole magnet [16]. Due to the bent beam trajectory, the horizontal angular divergence of the photon beam is 7 mrad, while the vertical divergence is assumed to be dominated by the natural divergence of the emitted photon beam. The photon beam travels through the photon beam pipe before impinging on the photoneutron target. In the following calculations, we have considered a distance of 70 m between the dipole and the target region, resulting in a photon beam of approximately $50 \text{ cm} \times 2 \text{ cm}$ at the target. The photon distribution has been assumed to be uniform over this area, while the photon energy was sampled from the characteristic synchrotron radiation spectrum, as shown in Fig. 2. The target materials for the sensitivity study were selected to cover the two primary categories of photoneutron converters. High-Z materials (Uranium, Tantalum, Tungsten) were chosen for their high photonuclear cross-sections, which peak in the hundreds of millibarns but have a relatively high reaction threshold of 5-8 MeV. Conversely, low-Z materials (Beryllium, Deuterium) were chosen for their very low reaction thresholds (1-2 MeV), despite having much smaller cross-sections in the millibarn range [15]. Finally, a mixed-oxide target ($\text{UO}_2 + \text{D}_2\text{O} + \text{BeO}$) was tested to investigate potential merging effects, where the different cross-section thresholds might combine to maximize the neutron yield due to the broad synchrotron photon spectrum. Details about the targets are reported in Table I.

Table I. Composition and properties of the simulated target materials.

Target	Isotope	Atomic fraction (%)	Density (g/cm^3)	Max length (cm)
UO_2	^{238}U	33.3	10.2	25
	^{16}O	66.7		
BeO	^9Be	50	3.02	100
	^{16}O	50		
D_2O	^2H	66.7	1.10	300
	^{16}O	33.3		
W	^{182}W	26.5	19.3	10
	^{183}W	14.3		
	^{184}W	30.6		
	^{186}W	28.4		
Ta	^{181}Ta	100	17	10
mix	^2H	25.0	4.77	60
	^9Be	12.5		
	^{16}O	50.0		
	^{238}U	12.5		

One of the key objectives of this study was to investigate the effect of the target transverse (XY) dimensions on the total photoneutron yield. To this end, two distinct geometric cases were simulated:

1. **Tailored Case:** The target XY dimensions (60 cm x 4 cm) are close to those of the photon source.
2. **Expanded Case:** The target XY dimensions (100 cm x 100 cm) are larger than the source.

This two-case approach was designed to account for the effect of non-first-flight photons. In fact, when a primary photon interacts with the target, the most probable reactions are photoatomic (namely, Compton scattering or pair production). The resulting secondary photons (from scattering) and bremsstrahlung photons (from the subsequent electron/positron transport) are emitted at different angles. The Ex-

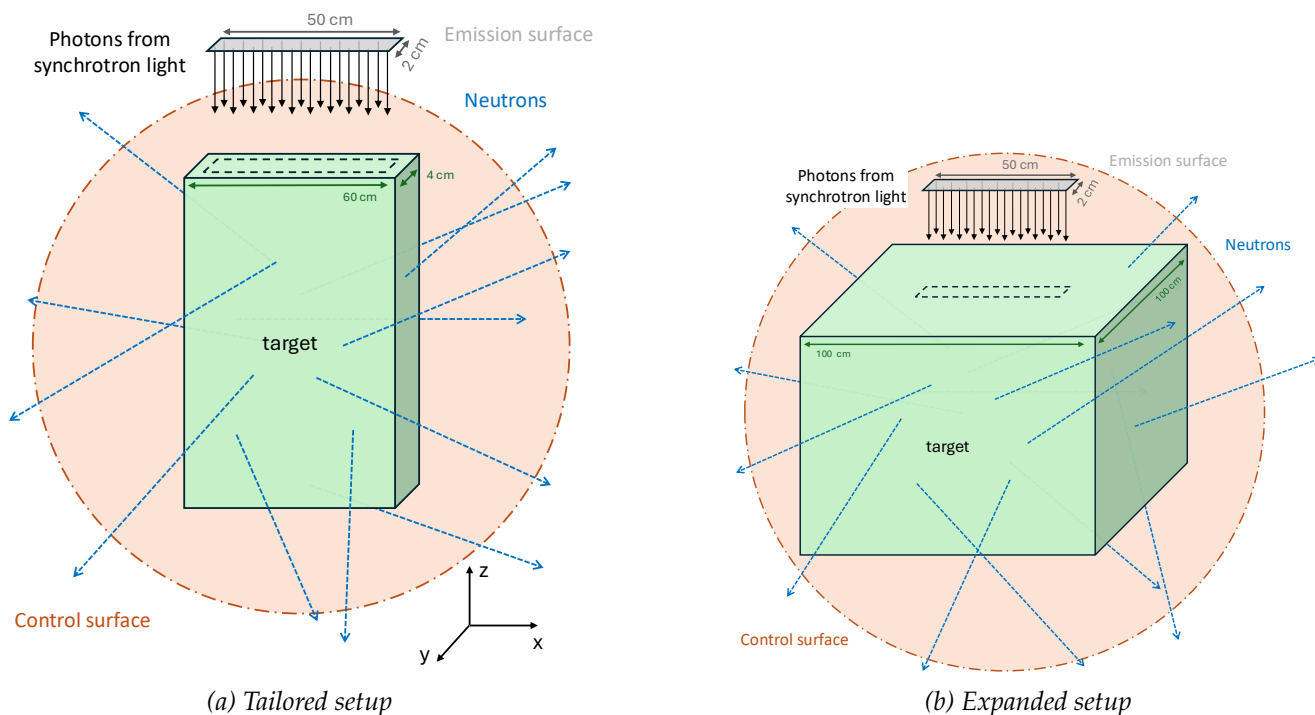


Figure 3. Simulation setup for the two considered cases.

panded Case is designed to capture these secondary, off-axis photons, which would otherwise escape a smaller target, thereby maximizing the interaction volume. However, this increase in transversal size presents a trade-off, as it also increases the probability of photoneutron auto-absorption (self-shielding) within the target, which can reduce the final measured yield.

The photoneutron yield was defined as the total number of neutrons crossing a large, spherical control surface (500 cm radius) surrounding the target, tallied per source particle. Another key analysis regarded the effect of the target length (i.e., the thickness along the Z-axis) for all target materials. The photoneutron yield was recorded for increasing target lengths in both the tailored and expanded configurations to determine the optimal design. Figure 3 reports the two simulation setups.

3. RESULTS

All simulation results were obtained from the non-official OpenMC branch developed by [11], that enabled the photonuclear interactions. Each OpenMC ran 3×10^6 source photons, leading to quantities with statistical uncertainty $\pm 0.5\%$. The primary output is the neutron current scored at the sphere boundary, expressed in terms of neutrons/source photon. This value represents the net photoneutron yield, accounting for neutrons produced by photonuclear reactions (the dominant component) and subsequent (n,xn) reactions, minus any neutrons absorbed within the target.

To provide physically meaningful comparisons, the Photoneutron Yield (PY) scored in the Monte Carlo calculation is evaluated in terms of neutrons/s, normalizing the current by the total synchrotron photon flux, assumed to be 2.01×10^{17} photons/s. This intensity corresponds to 200 kW of photon beam power. These values are a quantitative realistic estimation of the synchrotron capabilities. Figure 4 shows the photoneutron yield as a function of target length for all simulated materials in both the 'tailored' and 'expanded' configurations. This analysis identifies the optimal thickness for each material and geometric case. In all proposed tailored targets, a plateau in the PY is reached, meaning that all the useful photons have interacted with the target. The plateau value depends on the material's photonuclear mi-

cross-section and its density: for this reason, for each composition we decided to interrupt the simulation cases at a certain target length. The best performing material is UO_2 , as uranium can exploit multiple photoneutron production channels, including $(\gamma, 2n)$ and photofission. Furthermore, in all simulations, the 'expanded' scenario initially shows a maximum yield comparable to the 'tailored' case, but as the length increases, its PY begins to decrease due to the increased self-shielding of the neutrons produced. This self-shielding effect is observed in all cases except for heavy water (D_2O), which exhibits a very large photon mean free path [13] (30-60 cm, depending on photon energy) and a negligible neutron absorption cross-section.

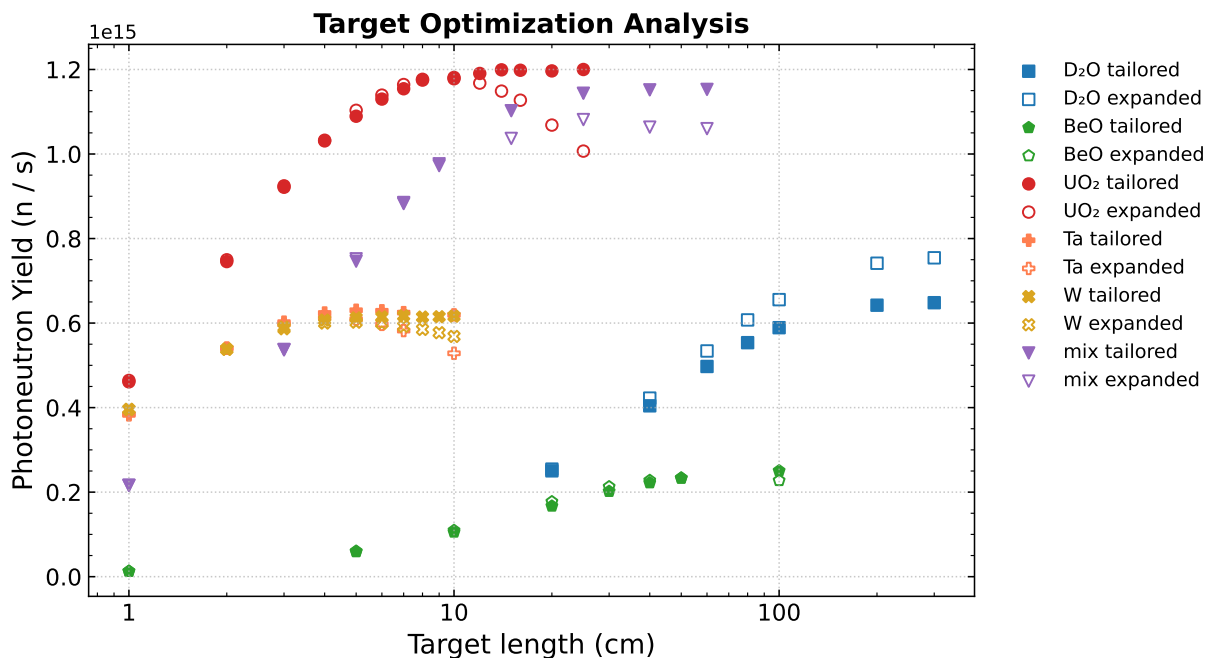


Figure 4. Photoneutron yield as a function of target length for all tested materials in both the 'tailored' and 'expanded' geometric cases.

The optimal photoneutron yield obtained from synchrotron radiation for each target material is then compared with published values for conventional LINAC-driven neutron sources, as reported in Table ???. The literature data originate from mixed sources: in some cases both the neutron yield (n/s) and the corresponding beam power are provided, while in others only the specific yield (n/s/kW) is available. To represent the capabilities of LINAC-based systems "at their best", all literature values are *renormalized* to a beam power of 100 kW, which is generally considered the upper practical limit for current accelerator technology. This comparison demonstrates the effectiveness of the synchrotron's characteristic relative to a bremsstrahlung-based neutron production. This result highlights the significant increase in total neutron flux achievable, reaching values up to 1.2×10^{15} n/s in the optimal UO_2 target. This demonstrates the huge synchrotron's potential for driving high-flux neutron applications like ADS.

4. CONCLUSIONS

This study demonstrates the strong potential of synchrotron radiation as a driver for high-flux photoneutron sources. Using detailed Monte Carlo simulations with photonuclear transport in OpenMC, the neutron yield across a broad set of materials and geometries was quantified. Uranium-based targets emerged as the most effective converters, benefiting from multiple photonuclear reaction channels and



Target	Reference	E_e MeV	LINAC (highest power) $\times 10^{14}$ n/s	Synchrotron (highest power) $\times 10^{14}$ n/s	Ratio -
D ₂ O	Dale 2005 [17]	10	0.715	6.482	9.066
D ₂ O	Fynan 2021 [13]	20	1.400	6.482	4.630
UO ₂	Mytsikov 2019 [5]	100	3.010	12.000	3.987
UO ₂	Verma 2023 [14]	100	3.320	12.000	3.614
Ta	Petwal 2007 [4]	100	2.340	6.303	2.694
Ta	Petwal 2007 [4]	150	3.140	6.303	2.007
W	Mytsikov 2019 [5]	100	1.880	6.170	3.282
W	Lai 2022 [2]	50	1.625	6.170	3.797
W	Verma 2023 [14]	100	2.246	6.170	2.747

Table II. Comparison of Neutron Yields: Synchrotron vs Literature, with neutron yields expressed in units of 10^{14} n/s.

achieving yields $> 1 \times 10^{15}$ n/s for a single 200 kW beamline. On the other hand, low-Z materials such as deuterium and beryllium, while exhibiting lower total yields, may offer favourable spectral and moderation characteristics that could support hybrid target designs. For all materials, increasing the target size for all the cases improves photon utilization up to a saturation point, beyond which self-shielding reduces efficiency. The advantages of a synchrotron driven neutron sources are preliminary pointed out in this work, namely the much higher total power deliverable by synchrotrons and the possibility of multiplexing dozens of independent beamlines around a single ring. This architecture could enable total neutron outputs approaching 10^{17} n/s across a full facility. These results establish a first quantitative foundation for synchrotron-driven photon sources and highlight their feasibility as scalable, high-yield alternatives to conventional electron-driven systems for advanced neutron science, materials testing, and accelerator-driven subcritical reactors.

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